

5CELEBAL DATA ENGINEERING INTERNSHIP

Name: Atharv Sakharam Patil

ID: CT_CSI_DE_1159

WEEK 4 Assignment

Objective: To understand Azure cloud concepts and build a complete data pipeline using Storage Account and Azure Data Factory.

Task 1: Explore Azure Portal & Create Resource Group Do:

- Explore Azure Portal
- Create a Resource Group

Deliverable: Screenshot of Resource Group

Created resource group 'celebal-internship-cei' in Central India under Azure for Students subscription.

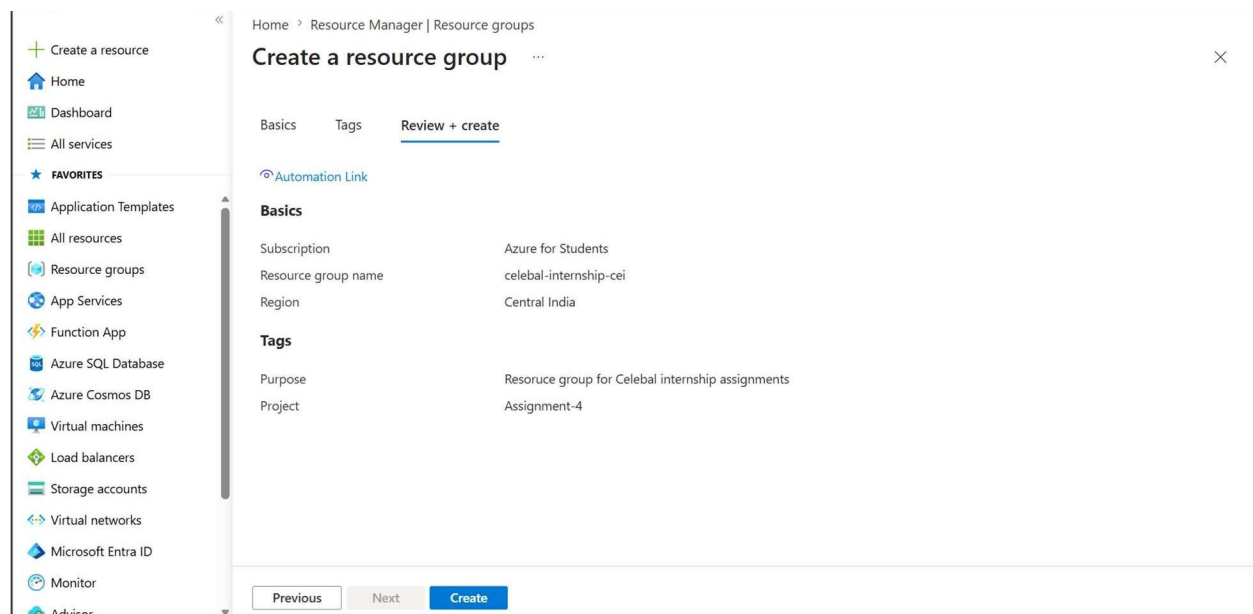


Figure: Resource Group creation review

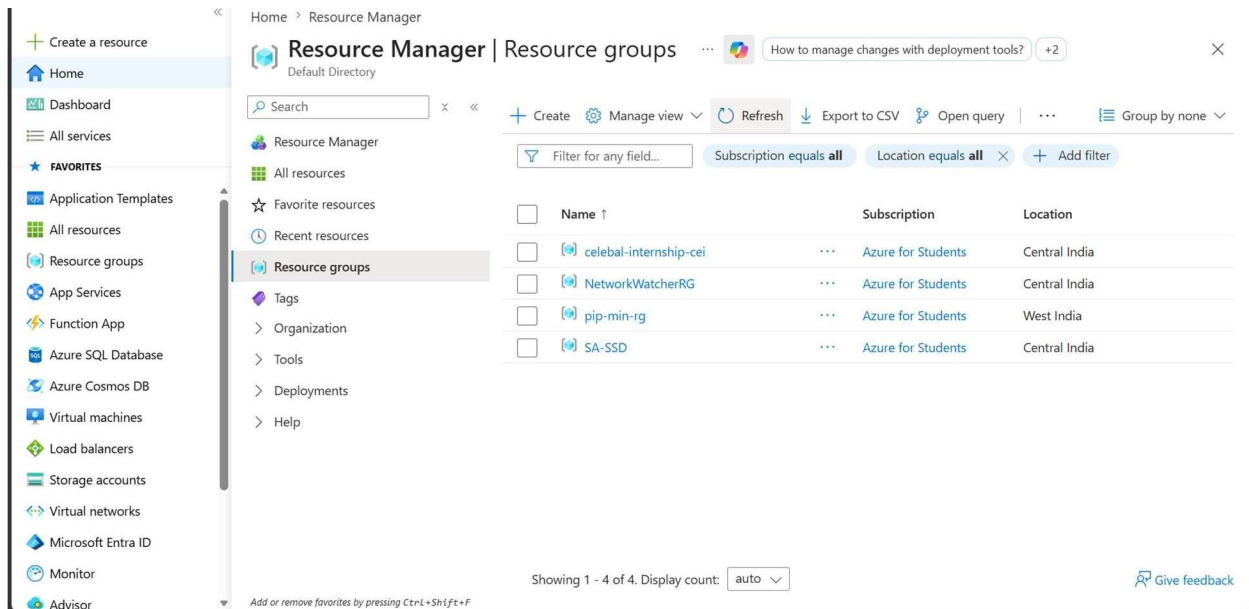


Figure: Resource Groups list

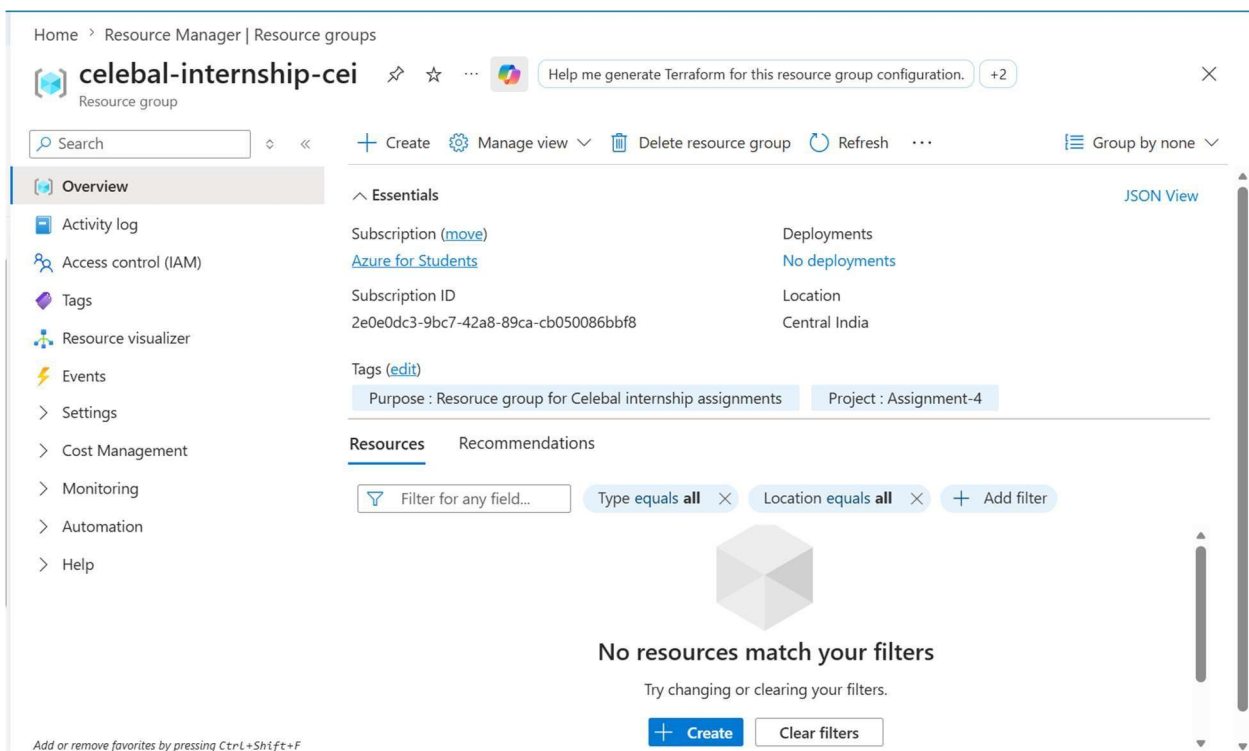


Figure: Created Resource Group overview

Task 2: Storage Setup Do:

- Create a Storage Account
- Create a Blob Container
- Upload a CSV file

Deliverable: Screenshot of container with uploaded CSV

Storage account and blob container were configured successfully. SampleSuperstore.csv uploaded.

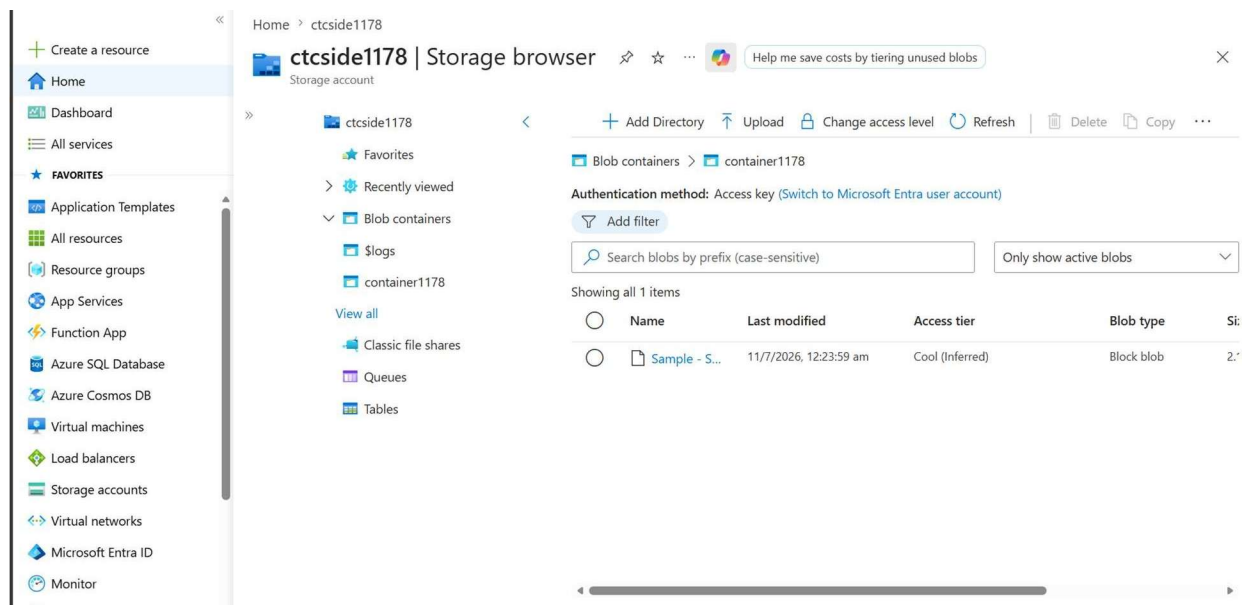


Figure: Blob container with uploaded CSV

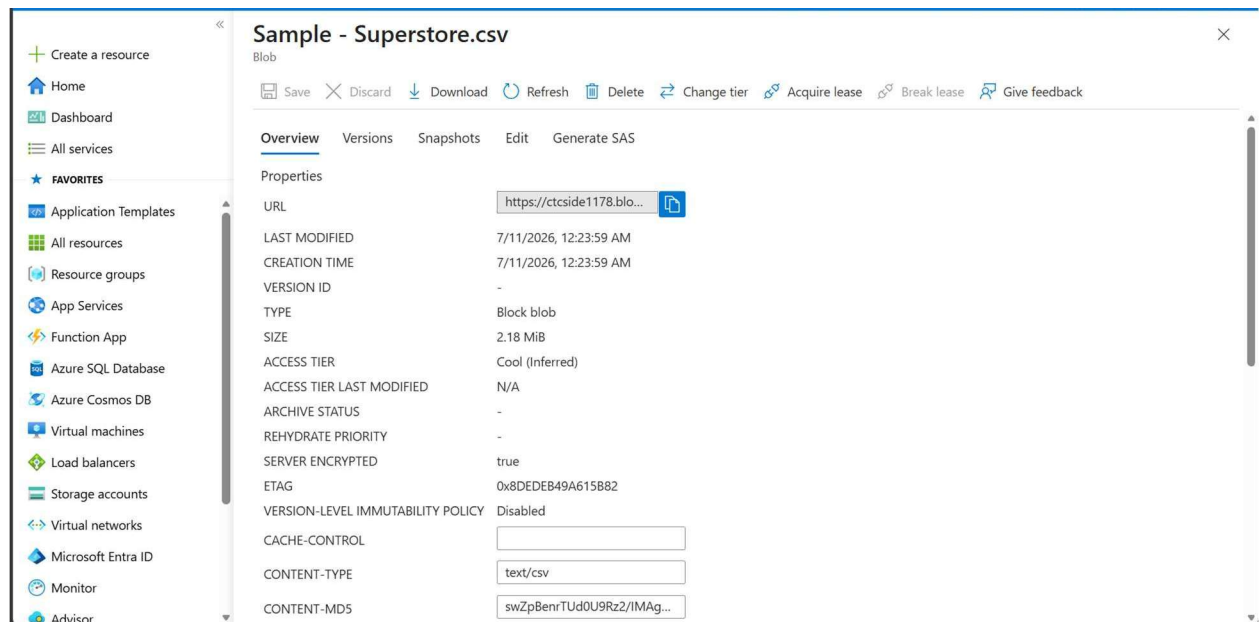


Figure: Uploaded CSV blob properties

Task 3: Azure Data Factory Basics Do:

- Create Azure Data Factory
- Explore Author, Monitor and Manage
- Create Linked Service
- Create Source and Destination Datasets
- Configure Get Metadata

Deliverable: Screenshots of Linked Service, Datasets and Get Metadata activity

Linked service AzureBlobStorage1 was created to connect ADF with Azure Blob Storage. Two datasets (source and destination) were configured. Get Metadata validates file existence, size and last modified time.

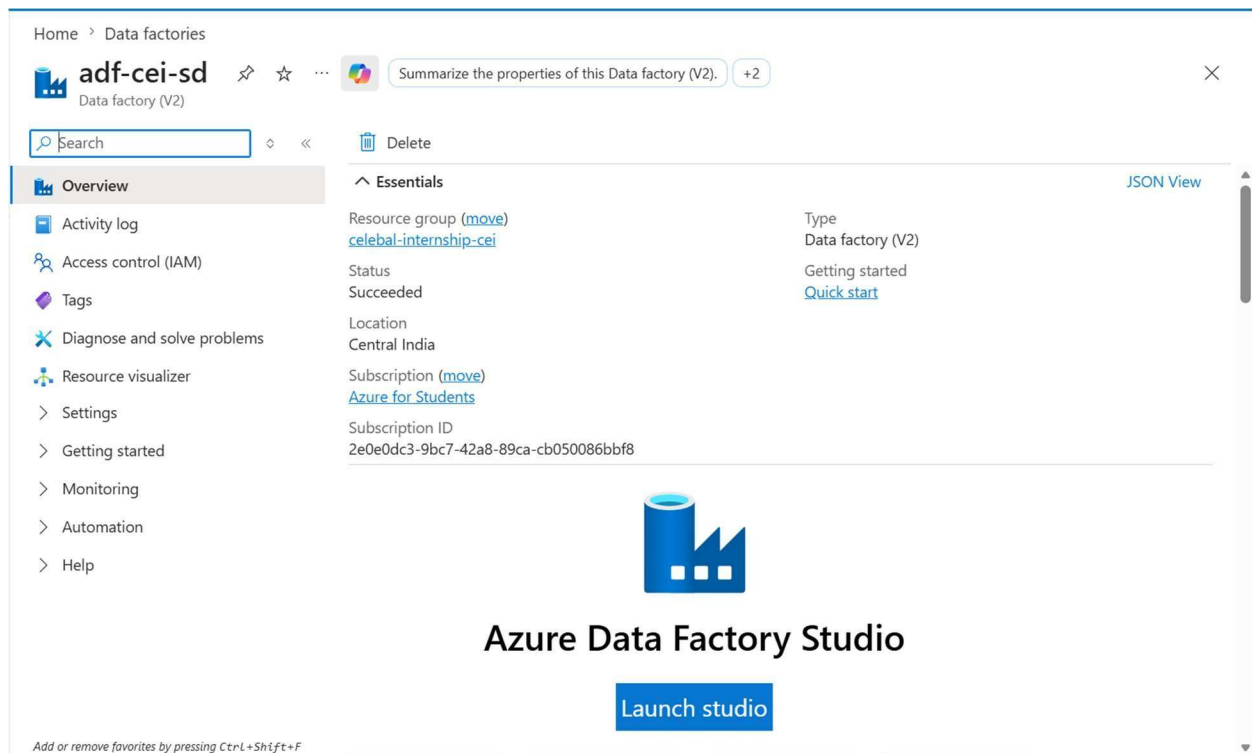


Figure: Azure Data Factory overview

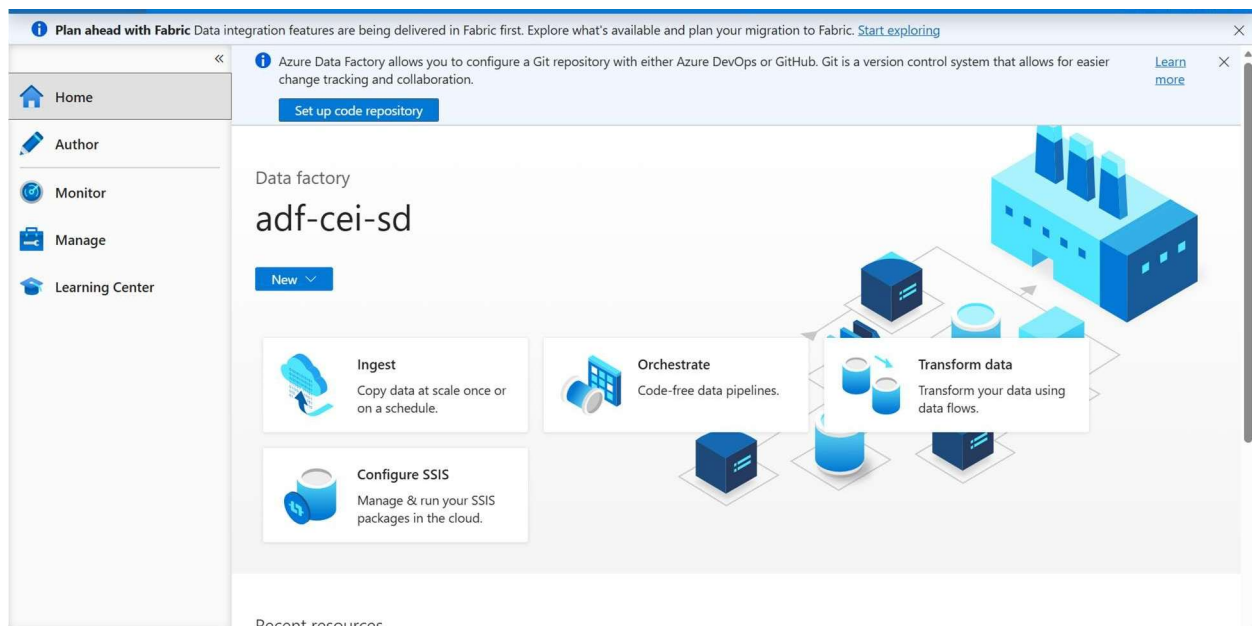


Figure: ADF Studio

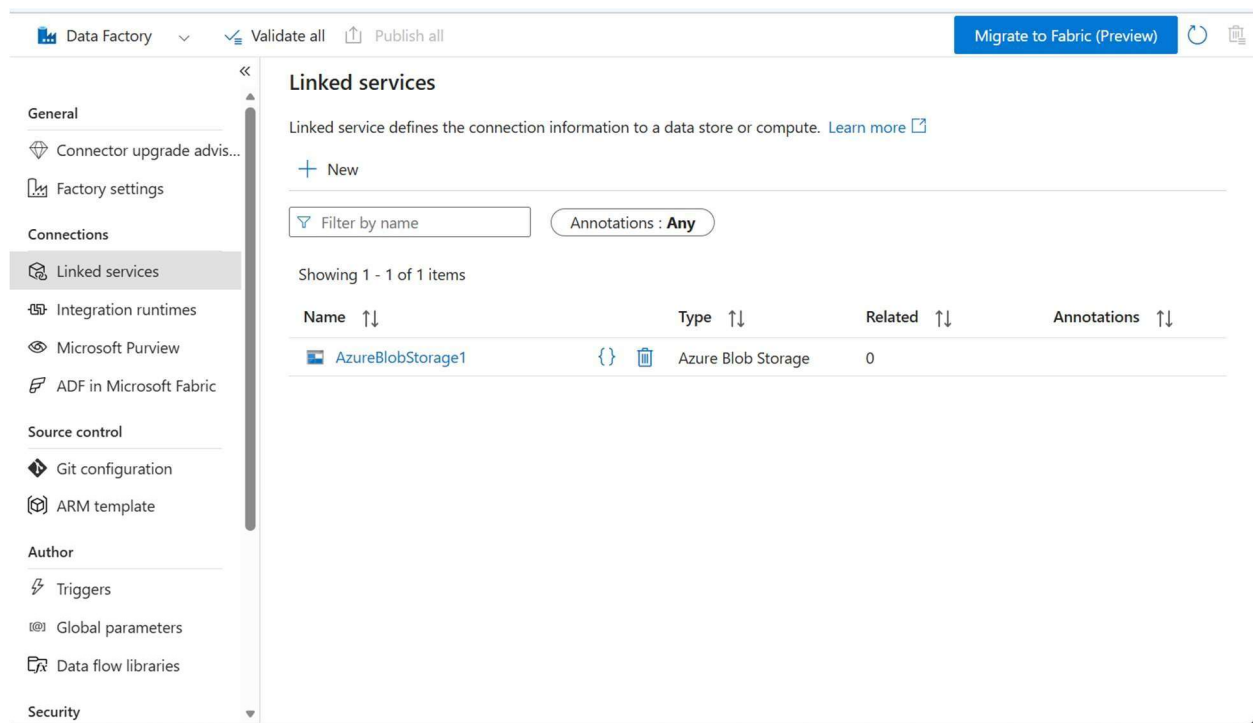


Figure: Linked Service



Figure: Source Dataset

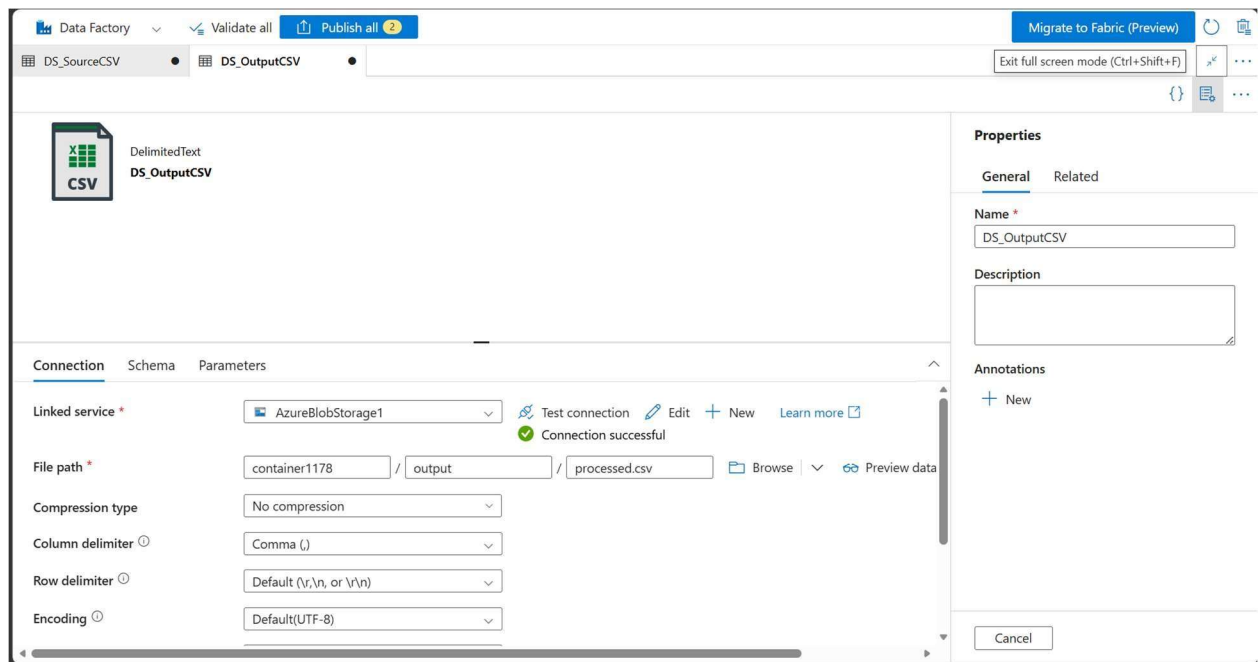


Figure: Destination Dataset

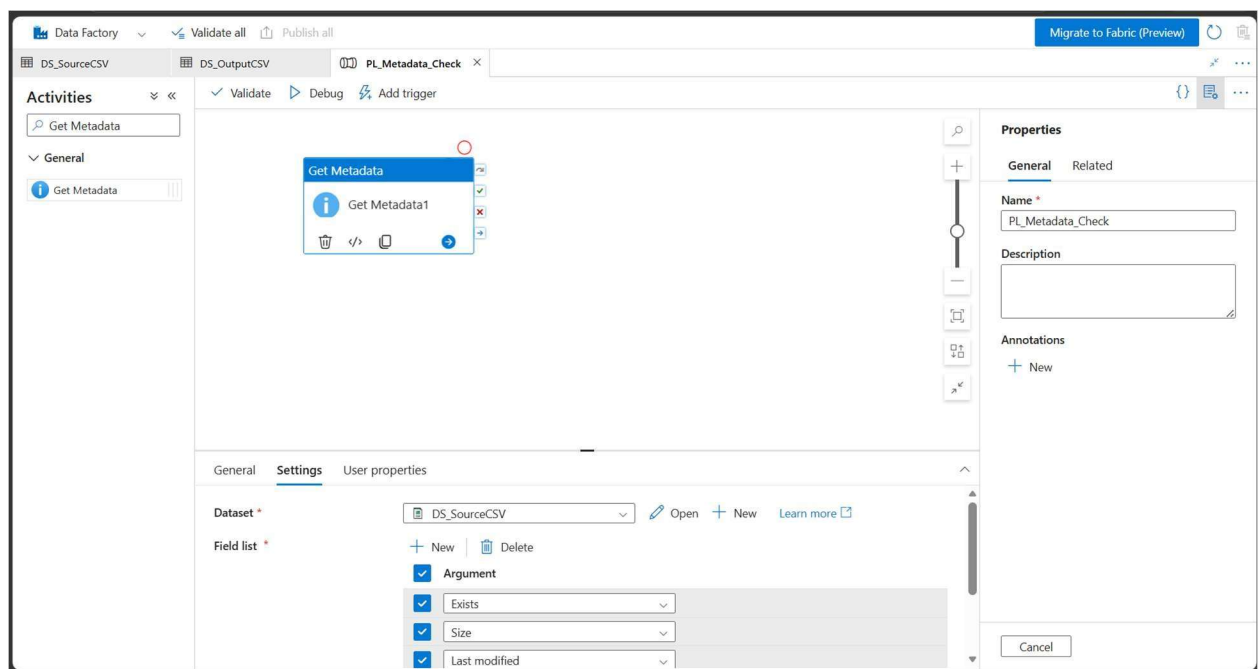


Figure: Get Metadata activity

Task 4: Pipeline Development Do:

- Create Copy Data pipeline
- Configure Source and Sink datasets
- Connect Get Metadata to Copy Data □ Validate pipeline

Deliverable: Screenshot of pipeline design

ForEach activity was optional and not required because only a single CSV file was processed.

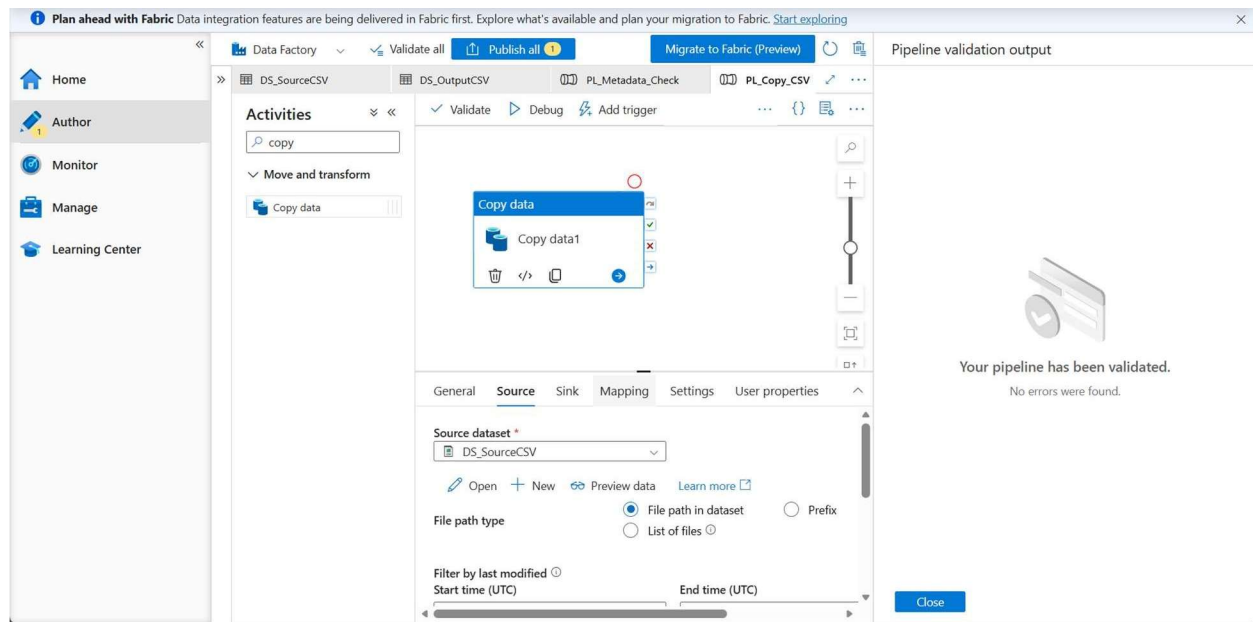


Figure: Copy Data activity configuration and design

Task 5: Pipeline Execution Do:

- Run pipeline using Debug
- Verify successful execution
- Validate copied output

Deliverable: Pipeline execution succeeded

Pipeline executed successfully through Debug mode. Metadata validation completed and the CSV file was copied to output/processed.csv.

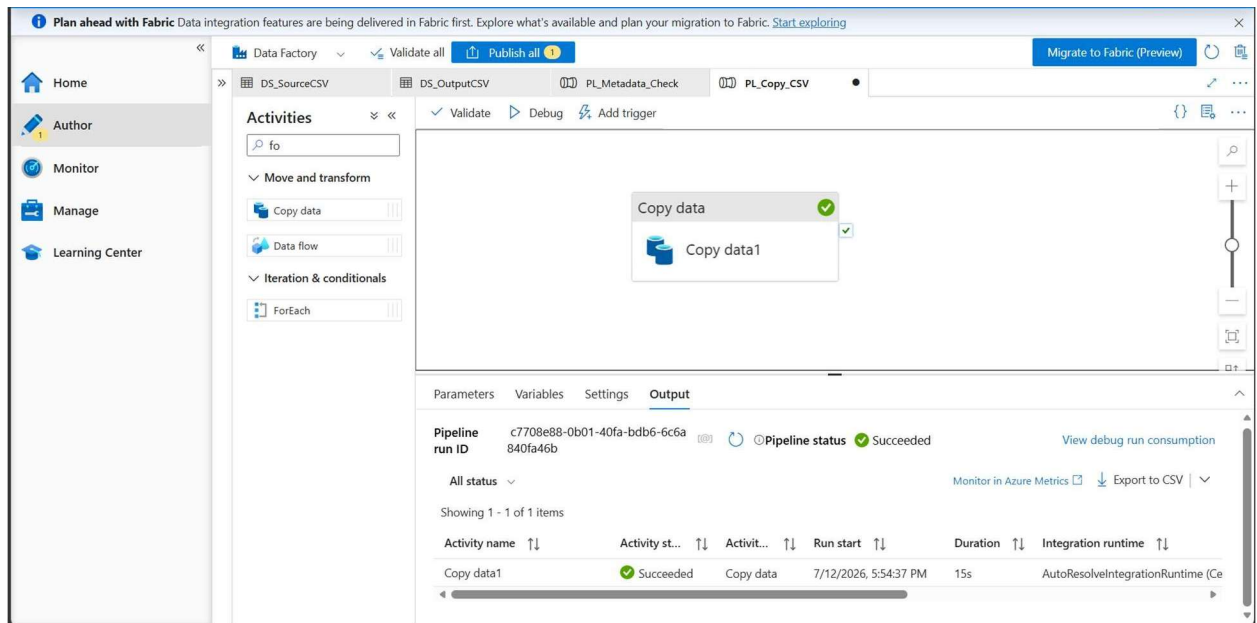


Figure: Copy Data succeeded

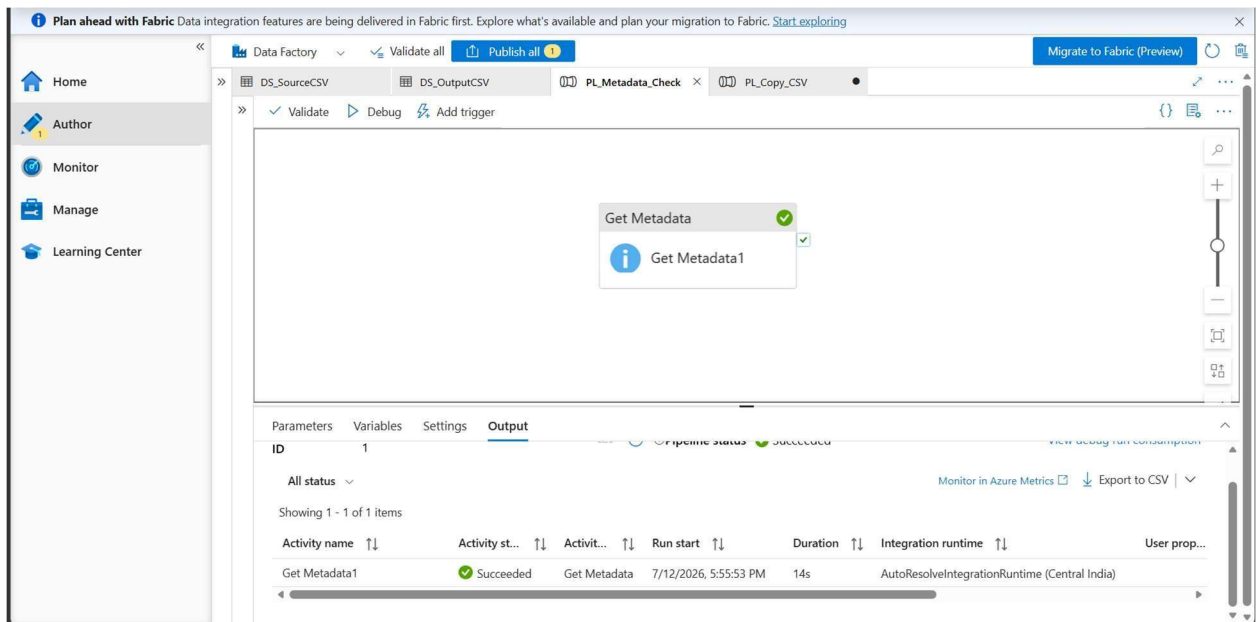


Figure: Get Metadata succeeded

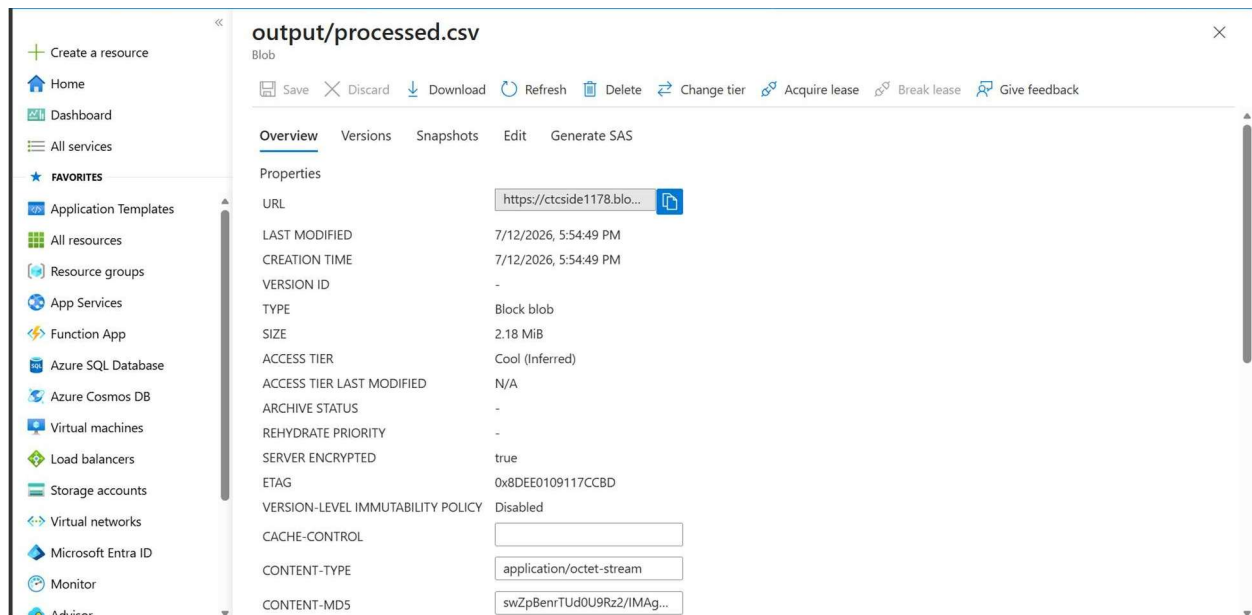


Figure: Processed CSV blob

The screenshot shows a Microsoft Excel spreadsheet with the following data:

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer	Customer Segment	Country	City	State	Postal Code	Region	Product ID	Category	Sub-Category	Product Name	Sales	Quantity	Discount
1	CA-2016-1	11/18/2015	11/18/2015	Standard Class	Sean O'Donoghue	Consumer	United States	Fort Lauderdale	Florida	33311	South	FUR-BO-1	Furniture	Bookcases	Bush Somerset	261.96	2	0
2	CA-2016-1	11/18/2015	11/18/2015	Standard Class	Claire Gutierrez	Consumer	United States	Henderson	Kentucky	42420	South	FUR-CH-1	Furniture	Chairs	Honolulu Deluxe	731.94	3	0
3	CA-2016-1	6/16/2016	6/16/2016	Standard Class	Darrin Van Dyke	Corporate	United States	Los Angeles	California	90036	West	OFF-LA-1C	Office Supplies	Labels	Self-Adhesive	14.62	2	0
4	US-2015-1	10/18/2015	10/18/2015	Standard Class	Sean O'Donoghue	Consumer	United States	Fort Lauderdale	Florida	33311	South	FUR-TA-1C	Furniture	Tables	Bretford Case	957.5775	5	0
5	US-2015-1	10/18/2015	10/18/2015	Standard Class	Sean O'Donoghue	Consumer	United States	Fort Lauderdale	Florida	33311	South	OFF-ST-10	Office Supplies	Storage	Eldon Foldout	22.368	2	0
6	CA-2014-1	6/14/2014	6/14/2014	Standard Class	Brosina Hc	Consumer	United States	Los Angeles	California	90032	West	FUR-FU-1C	Furniture	Furnishings	Eldon Express	48.86	7	0
7	CA-2014-1	6/14/2014	6/14/2014	Standard Class	Brosina Hc	Consumer	United States	Los Angeles	California	90032	West	OFF-AR-1C	Office Supplies	Art	Newell 322	7.28	4	0
8	CA-2014-1	6/14/2014	6/14/2014	Standard Class	Brosina Hc	Consumer	United States	Los Angeles	California	90032	West	TEC-PH-1C	Technology	Phones	Mitel 5320	907.152	6	0
9	CA-2014-1	6/14/2014	6/14/2014	Standard Class	Brosina Hc	Consumer	United States	Los Angeles	California	90032	West	OFF-BI-10	Office Supplies	Binders	DXL Angle	18.504	3	0
10	CA-2014-1	6/14/2014	6/14/2014	Standard Class	Brosina Hc	Consumer	United States	Los Angeles	California	90032	West	OFF-AP-1C	Office Supplies	Appliances	Belkin F5C	114.9	5	0
11	CA-2014-1	6/14/2014	6/14/2014	Standard Class	Brosina Hc	Consumer	United States	Los Angeles	California	90032	West	FUR-TA-1C	Furniture	Tables	Chromcraft	1706.184	9	0
12	CA-2014-1	6/14/2014	6/14/2014	Standard Class	Brosina Hc	Consumer	United States	Los Angeles	California	90032	West	TEC-PH-1C	Technology	Phones	Konftel 250	911.424	4	0
13	CA-2017-1	4/15/2017	4/20/2017	Standard Class	Andrew All	Consumer	United States	Concord	North Carolina	28027	South	OFF-PA-1C	Office Supplies	Paper	Xerox 1967	15.552	3	0
14	CA-2016-1	11/18/2015	11/18/2015	Standard Class	Irene Madrid	Consumer	United States	Seattle	Washington	98103	West	OFF-BI-10	Office Supplies	Binders	Fellowes P	407.976	3	0
15	US-2015-1	11/22/2011	11/26/2011	Standard Class	Harold Pav	Home Office	United States	Fort Worth	Texas	76106	Central	OFF-AP-1C	Office Supplies	Appliances	Holmes Re	68.81	5	0
16	US-2015-1	11/22/2011	11/26/2011	Standard Class	Harold Pav	Home Office	United States	Fort Worth	Texas	76106	Central	OFF-BI-10	Office Supplies	Binders	Storex Dur	2.544	3	0
17	CA-2014-1	11/18/2015	11/18/2015	Standard Class	Pete Kriz	Consumer	United States	Madison	Wisconsin	53711	Central	OFF-ST-10	Office Supplies	Storage	Stur-D-Sto	665.88	6	0
18	CA-2014-1	5/15/2014	5/15/2014	Standard Class	Alejandro I	Consumer	United States	West Jordan	Utah	84084	West	OFF-ST-10	Office Supplies	Storage	Fellowes S	55.5	2	0
19	CA-2014-1	18/27/2014	18/27/2014	Standard Class	Zuschuss I	Consumer	United States	San Francisco	California	94109	West	OFF-AR-1C	Office Supplies	Art	Newell 341	8.56	2	0
20	CA-2014-1	18/27/2014	18/27/2014	Standard Class	Zuschuss I	Consumer	United States	San Francisco	California	94109	West	TEC-PH-1C	Technology	Phones	Cisco SPA	213.48	3	0
21	CA-2014-1	18/27/2014	18/27/2014	Standard Class	Zuschuss I	Consumer	United States	San Francisco	California	94109	West	OFF-BI-10	Office Supplies	Binders	Wilson Jon	22.72	4	0

Figure: Original CSV preview

AutoSave

Off

Save

Undo

Redo

process... Saved to this PC v

Search

File

Home

Insert

Draw

Page Layout

Formulas

Data

Review

View

Help

Acrobat

Clipboard

Font

Alignment

Number

Styles

Cells

Editing

Add-ins

Add-ins

Comments

Share v

S1

Quantity

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
1	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer	Customer Segment	Country	City	State	Postal Code	Region	Product ID	Category	Sub-Category	Product Name	Sales	Quantity		
2	1	CA-2016-1	6/16/2016	6/16/2016	Standard	Second Cl	CG-12520	Claire Gut	Consumer	United Sta	Henderson	Kentucky	42420	South	FUR-BO-1	Furniture	Bookcase: Bush Soma	261.96	2	
3	2	CA-2016-1	6/16/2016	6/16/2016	Standard	Second Cl	CG-12520	Claire Gut	Consumer	United Sta	Henderson	Kentucky	42420	South	FUR-CH-1	Furniture	Chairs Hon Delux	731.94	3	
4	3	CA-2016-1	6/16/2016	6/16/2016	Standard	Second Cl	DV-13045	Darrin Van	Corporate	United Sta	Los Angeles	California	90036	West	OFF-LA-1	Office Sup	Labels Self-Adhes	14.62	2	
5	4	US-2015-1	10/18/201	10/18/201	Standard	CSO-20335	Sean O'Do	Consumer	United Sta	Fort Laude	Florida	33311	South	FUR-TA-1	Furniture	Tables Brettford C	957.5775	5		
6	5	US-2015-1	10/18/201	10/18/201	Standard	CSO-20335	Sean O'Do	Consumer	United Sta	Fort Laude	Florida	33311	South	OFF-ST-10	Office Sup	Storage Eldon Fold	22.368	2		
7	6	CA-2014-1	6/14/2014	6/14/2014	Standard	CBH-11710	Brosina Hc	Consumer	United Sta	Los Angeles	California	90032	West	FUR-FU-1	Furniture	Furnishing Eldon Expr	48.86	7		
8	7	CA-2014-1	6/14/2014	6/14/2014	Standard	CBH-11710	Brosina Hc	Consumer	United Sta	Los Angeles	California	90032	West	OFF-AR-1	Office Sup	Art Newell 322	7.28	4		
9	8	CA-2014-1	6/14/2014	6/14/2014	Standard	CBH-11710	Brosina Hc	Consumer	United Sta	Los Angeles	California	90032	West	TEC-PH-1	Technolog	Phones Mitel 5320	907.152	6		
10	9	CA-2014-1	6/14/2014	6/14/2014	Standard	CBH-11710	Brosina Hc	Consumer	United Sta	Los Angeles	California	90032	West	OFF-BI-10	Office Sup	Binders DXL Angle	18.504	3		
11	10	CA-2014-1	6/14/2014	6/14/2014	Standard	CBH-11710	Brosina Hc	Consumer	United Sta	Los Angeles	California	90032	West	OFF-AP-1	Office Sup	Appliance: Belkin FSC	114.9	5		
12	11	CA-2014-1	6/14/2014	6/14/2014	Standard	CBH-11710	Brosina Hc	Consumer	United Sta	Los Angeles	California	90032	West	FUR-TA-1	Furniture	Tables Chromcral	1706.184	9		
13	12	CA-2014-1	6/14/2014	6/14/2014	Standard	CBH-11710	Brosina Hc	Consumer	United Sta	Los Angeles	California	90032	West	TEC-PH-1	Technolog	Phones Konftel 25i	911.424	4		
14	13	CA-2017-1	4/15/2017	4/15/2017	Standard	CAA-10480	Andrew All	Consumer	United Sta	Concord	North Car	28027	South	OFF-PA-1	Office Sup	Paper Xerox 196	15.552	3		
15	14	CA-2016-1	6/16/2016	6/16/2016	Standard	CI-15070	Irene Madr	Consumer	United Sta	Seattle	Washingto	98103	West	OFF-BI-10	Office Sup	Binders Fellowes P	407.976	3		
16	15	US-2015-1	11/22/201	11/22/201	Standard	CHP-14815	Harold Pav	Home Offi	United Sta	Fort Worth	Texas	76106	Central	OFF-AP-1	Office Sup	Appliance: Holmes Re	68.81	5		
17	16	US-2015-1	11/22/201	11/22/201	Standard	CHP-14815	Harold Pav	Home Offi	United Sta	Fort Worth	Texas	76106	Central	OFF-BI-10	Office Sup	Binders Storex Dur	2.544	3		
18	17	CA-2014-1	11/18/201	11/18/201	Standard	CPK-19075	Pete Kriz	Consumer	United Sta	Madison	Wisconsin	53711	Central	OFF-ST-10	Office Sup	Storage Stur-D-Sto	665.88	6		
19	18	CA-2014-1	15/13/2014	15/13/2014	Standard	CI-AG-10270	Alejandro I	Consumer	United Sta	West Jord	Utah	84084	West	OFF-ST-10	Office Sup	Storage Fellowes S	55.5	2		
20	19	CA-2014-1	18/27/2014	18/27/2014	Standard	CI-ZD-21925	Zuschuss I	Consumer	United Sta	San Franci	California	94109	West	OFF-AR-1	Office Sup	Art Newell 34i	8.56	2		
21	20	CA-2014-1	18/27/2014	18/27/2014	Standard	CI-ZD-21925	Zuschuss I	Consumer	United Sta	San Franci	California	94109	West	TEC-PH-1	Technolog	Phones Cisco SPA	213.48	3		
22	21	CA-2014-1	18/27/2014	18/27/2014	Standard	CI-ZD-21925	Zuschuss I	Consumer	United Sta	San Franci	California	94109	West	OFF-BI-10	Office Sup	Binders Wilson Jon	22.72	4		

processed

+

processed

Ready

Scroll Lock

Accessibility: Unavailable

Average: 3.789573744

Count: 9995

Sum: 37873

100%

Figure: Processed CSV preview

Task 6: IAM Roles Do:

- Assign Reader role
- Assign Contributor/Data Factory Contributor role
- Grant Storage Blob Data Contributor access to ADF managed identity

Deliverable: Screenshot of role assignments

ADF managed identity was granted Storage Blob Data Contributor access to interact with Azure Blob Storage securely.

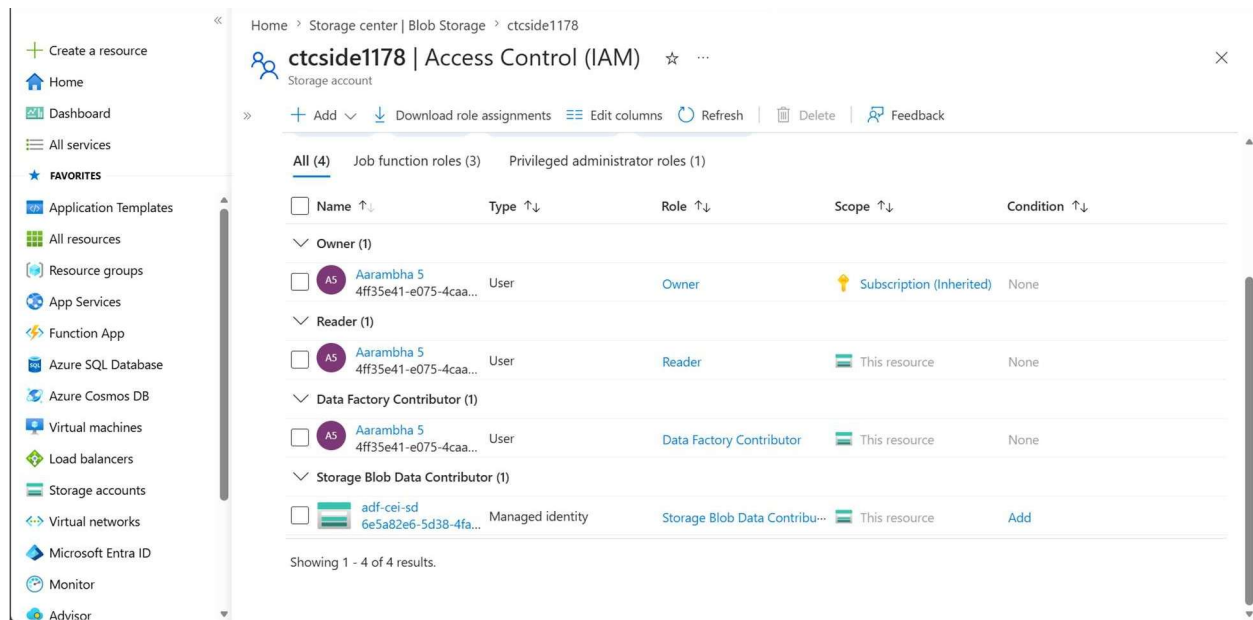


Figure: Storage Blob Data Contributor assigned to ADF managed identity

Mini Project (Final Task)

Problem Statement: Build a complete pipeline that reads a CSV file from Azure Blob Storage, validates metadata using Get Metadata, and copies the CSV to a new location in Blob Storage.

Implementation Summary

- Source: Sample-Superstore.csv stored in Azure Blob Storage.
- Linked Service: AzureBlobStorage1.
- Datasets: DS_SourceCSV and DS_OutputCSV.
- Metadata Validation: Get Metadata activity checks file existence, size and last modified date.
- Pipeline: PL_CSV_DataPipeline executes Get Metadata followed by Copy Data.
- Destination: output/processed.csv in the same blob container.
- Result: Pipeline executed successfully and destination file matched the source data.

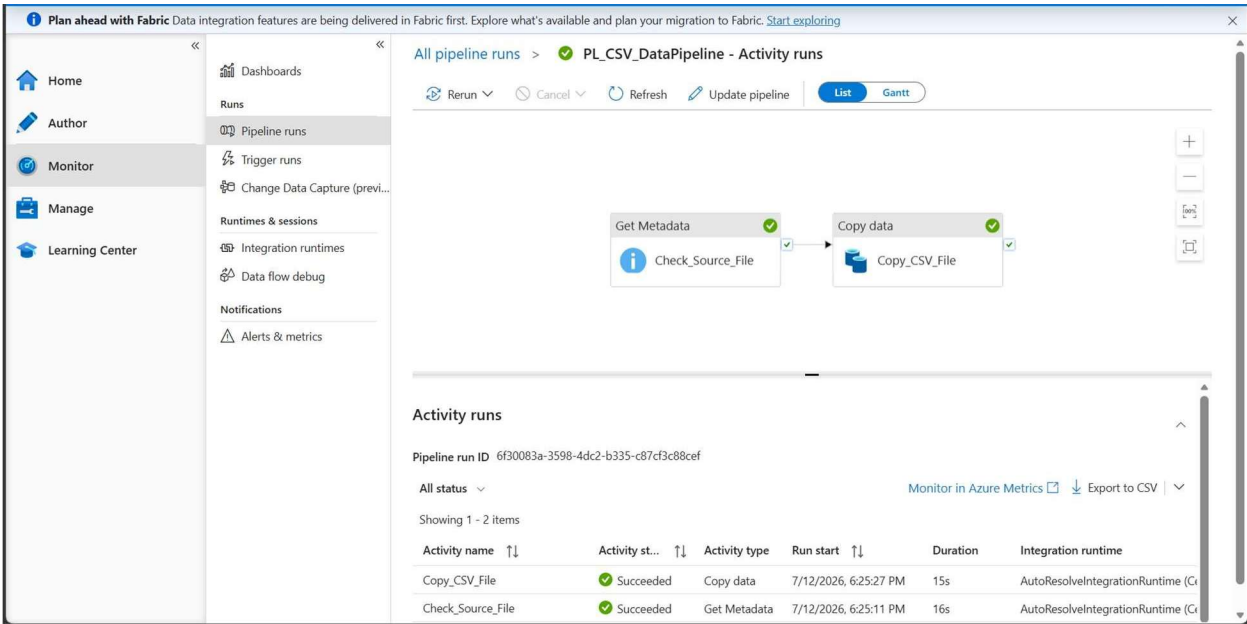


Figure: Final Mini Project Pipeline Execution

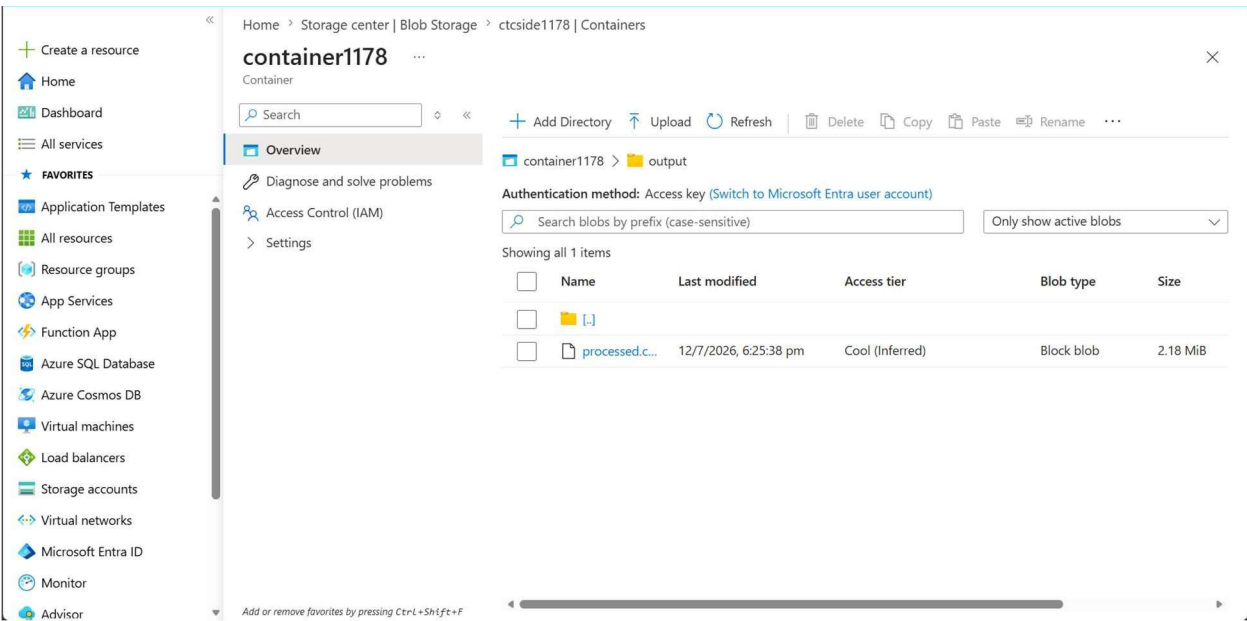


Figure: Data copied to destination

Input given:

```
{
  "dataset": {
    "referenceName": "DS_SourceCSV",
    "type": "DatasetReference",
    "parameters": {}
  },
  "fieldList": [
    "exists",
    "size",
    "lastModified"
  ],
  "storeSettings": {
    "type": "AzureBlobStorageReadSettings",
    "enablePartitionDiscovery": false
  },
  "formatSettings": {
    "type": "DelimitedTextReadSettings"
  }
}
```

Output

Copy to clipboard

```
{
  "exists": true,
  "size": 2287806,
  "lastModified": "2026-07-10T18:53:59Z",
  "effectiveIntegrationRuntime":
"AutoResolveIntegrationRuntime (Central India)",
  "executionDuration": 0,
  "durationInQueue": {
    "integrationRuntimeQueue": 11
  }
}
```

Monitor in Azure Metrics

Export to CSV

	Run start	Duration	Integration runtime
	7/12/2026, 6:25:27 PM	15s	AutoResolveIntegrationRuntime (C
Check_Source_File	7/12/2026, 6:25:11 PM	16s	AutoResolveIntegrationRuntime (C

Figure: Meta Data Verified